Dementia and its treatments

UoM Psychology Undergraduate Lecture — April 2026
(PowerPoint handout)

Dr. Ross A. Dunne

Outline

- 20 minutes on “Dementia”: a modern framework for understanding
- 20 minutes on the development of symptomatic treatments
- 5 minute break
- 20 minutes on disease-modifying treatments
- 5 minutes for questions on the material
- 10–20 minutes for Q&A with our trial participant
- 10 minutes for student questions

Part 1 — Dementia: a modern framework

This part covers the terminology, the biological versus clinical definition of Alzheimer's disease, the AT(N)(X) biomarker framework, and the clinical assessment workflow.

Audience poll

The interactive Slido poll used during the live lecture is not reproduced here. Students were invited to scan a QR code to join.

Alzheimer’s disease is not synonymous with dementia

Dementia is an umbrella term used to describe a state where an individual has lost the ability to carry out daily life activities independently because of cognitive impairment.

There are many types and causes of dementia. The most common ones are:

1. Alzheimer’s disease (AD) — pathological definition
2. Vascular dementia — aetiological definition
3. Dementia with Lewy bodies — molecular definition
4. Frontotemporal dementia — anatomical definition

Nature Neuroscience overview

A short Nature Neuroscience overview video was shown at this point.

Video: https://youtu.be/zTd0-A5yDZI

Alzheimer's disease is a disease like any other

- AD refers to the abnormal presence of Aβ and tau proteins which define AD among many other neurodegenerative diseases
- Some individuals may have dementia without Aβ or tau pathology
- Others (rarely) may have no clinical symptoms even in the presence of Aβ pathology

Alzheimer’s disease timeline

Timeline of Alzheimer’s disease pathology and clinical milestones, from Alzheimer’s 1907 case description through to disease-modifying antibody approvals in 2023–2024.

Biological vs clinical definition

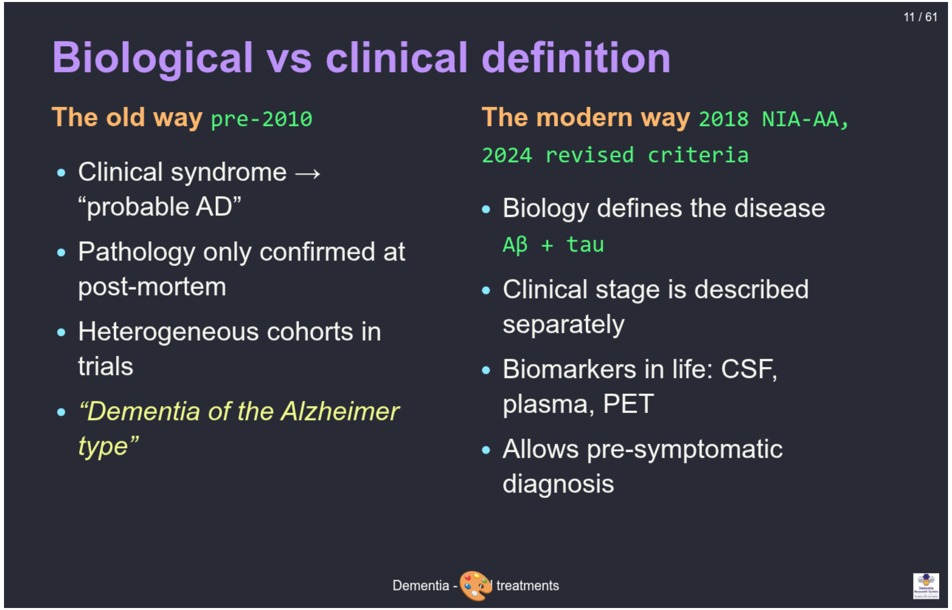
The old way (pre-2010)

- Clinical syndrome → "probable AD"
- Pathology only confirmed at post-mortem
- Heterogeneous cohorts in trials
- "Dementia of the Alzheimer type"

The modern way (2018 NIA-AA, 2024 revised criteria)

- Biology defines the disease (Aβ + tau)
- Clinical stage is described separately
- Biomarkers in life: CSF, plasma, PET
- Allows pre-symptomatic diagnosis

The biomarker cascade



Jack 2010/2013 biomarker cascade model: Aβ becomes abnormal earliest, then tau, neurodegeneration, cognition, and finally function. Pathology precedes symptoms by 15–20 years.

Audience poll: higher or lower?

Interactive Slido poll on prevalence estimates. Not reproduced in the handout.

The AT(N)(X) framework

The modern biomarker classification, used to characterise patients independently of clinical stage. Each axis captures a different category of pathology.

AT(N)(X): A — Amyloid

What it measures: cerebral Aβ pathology.

- CSF Aβ42/40 ratio (low = abnormal)
- Plasma Aβ42/40 (replaces some lumbar punctures)
- Amyloid-PET (florbetapir, flutemetamol)

AT(N)(X): T — Tau

What it measures: tau pathology.

- CSF p-tau181, p-tau217
- **Plasma p-tau217**
- Tau-PET (flortaucipir, tracks Braak staging)

AT(N)(X): N — Neurodegeneration

What it measures: downstream injury, not specific to AD.

- Structural MRI atrophy (MTA score)
- FDG-PET (temporo-parietal hypometabolism)
- CSF total tau

AT(N)(X): X — Other

What it measures: vascular, inflammatory, and synaptic axes added in 2024.

- Vascular markers (WMH, infarcts)
- Inflammatory (GFAP, astrocyte activation)
- Synaptic (neurogranin, SNAP-25)

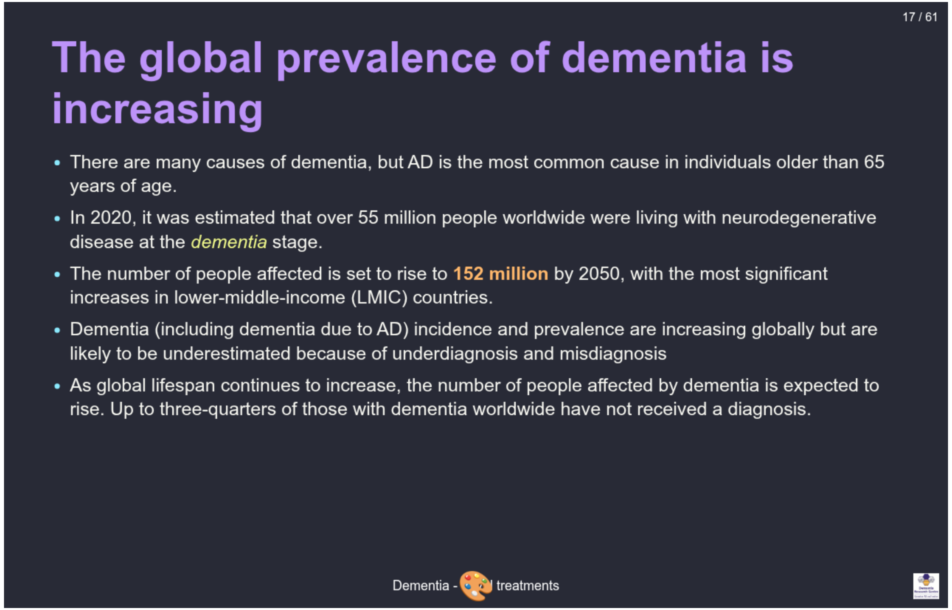
Audience poll: question 3

Interactive Slido poll. Not reproduced in the handout.

The global prevalence of dementia is increasing

- AD is the most common cause of dementia in individuals older than 65 years of age
- In 2020, over 55 million people worldwide were living with neurodegenerative disease at the *dementia* stage
- The number is set to rise to **139 million by 2050**, with the most significant increases in lower-middle-income (LMIC) countries
- Prevalence is likely underestimated because of underdiagnosis and misdiagnosis
- Up to three-quarters of those with dementia worldwide have not received a diagnosis

57.4M to 153M by 2050



Global dementia prevalence 2020 (~55–57M) versus projected 2050 figure (~139–153M), with the largest growth in LMIC countries.

Regions of greatest change

The largest projected increases are in:

- North Africa and the Middle East
- Sub-Saharan Africa
- South Asia
- Latin America

This reflects ageing populations in regions with limited diagnostic infrastructure.

Historical development of AD diagnostic criteria

- The 2023 revised criteria emphasises that AD is defined by its biology
- Only biomarkers proven to be accurate against an accepted reference standard should be used for diagnosis
- Biomarker tests should always be interpreted with clinical judgment

The clinician's expertise is crucial in three situations:

1. When the biomarker test result contradicts the clinical presentation
2. When evaluating the probable contribution of AD versus other pathologies
3. When assessing how other medical conditions could impact the biomarker results

AD diagnostic biomarkers

Core 1 (currently accepted for diagnosis)

- CSF Aβ42/40
- CSF p-tau181/Aβ42
- CSF t-tau/Aβ42
- Accurate plasma assays
- Amyloid-PET

Core 2 (emerging / research-grade)

- Plasma p-tau217 (near-CSF accuracy)
- Plasma MTBR-tau243
- GFAP (astrocyte activation)
- NfL (non-specific neurodegeneration)
- Tau-PET (Braak staging in life)

Differential diagnosis

In the rest of this section: the four most common dementia subtypes side by side. The handout shows one slide per subtype.

Differential diagnosis: AD

- **Onset**: insidious, episodic memory first
- **Signature**: temporo-parietal cognition; language; getting lost
- **Imaging**: medial temporal and parietal atrophy
- **Treatment cue**: AChEi and memantine; now anti-amyloid eligible

Differential diagnosis: DLB

- **Onset**: fluctuating cognition, visual hallucinations, REM sleep behaviour disorder, parkinsonism
- **Signature**: attention/executive > memory early; very neuroleptic-sensitive
- **Imaging**: relatively preserved MTL; DAT scan abnormal
- **Treatment cue**: rivastigmine often helpful; **avoid antipsychotics**

Differential diagnosis: FTD

- **Onset**: typically <65 y; behavioural variant or primary progressive aphasia
- **Signature**: disinhibition, apathy, loss of empathy, dietary change (bvFTD)
- **Imaging**: frontal / anterior temporal atrophy
- **Treatment cue**: AChEi unhelpful or may worsen behaviour; SSRIs and environmental management

Differential diagnosis: VaD

- **Onset**: stepwise, after vascular events
- **Signature**: executive dysfunction, slowed processing, focal neurology
- **Imaging**: infarcts, white matter hyperintensities, microbleeds
- **Treatment cue**: aggressive vascular risk control; modest AChEi benefit

Reversible mimics

Always rule out treatable contributors to cognitive impairment:

- B12 / folate deficiency
- Hypothyroidism
- Depression (pseudo-dementia)
- Normal pressure hydrocephalus (gait, urinary, cognitive)
- Drug effects (anticholinergics, benzodiazepines, opioids)
- Delirium (often superimposed)
- Obstructive sleep apnoea
- Subdural haematoma
- Alcohol misuse

Assessment

The full assessment in the live deck is a tabset; the handout shows one slide per assessment domain.

Assessment: Clinical history

Medical history from the patient and collateral history from an appropriate informant.

Background factors

- Educational attainment — quantitative and qualitative
- Occupational attainment — manual / skilled / professional
- Head injury or contact sports (rugby, boxing)
- Lifetime alcohol and substance use — peak and sustained
- Smoking history

Cognitive features

- Rapid forgetfulness, repetitive questioning
- Loss of recent memory with relative preservation of remote
- Functional decline (IADLs first, then ADLs)
- BPSD (agitation, depression, sleep)

Assessment: Cognitive examination

A cognitive, functional, behavioural and mental state examination, preferably using validated assessment scales.

- RBANS — sensitive to different sorts of decline
- Addenbrookes' Cognitive Evaluation (III) — used in 90% of UK memory clinics
- Mini Mental State Examination — insensitive to early decline
- Montreal Cognitive Assessment — better than MMSE, retains ceiling effects

Assessment: Examination

Physical examination, laboratory tests (to exclude comorbid conditions that may be impairing cognitive function), and review of medication (to identify drugs that may be impairing cognitive function).

Assessment: Neuroimaging

- **Structural MRI** (first-line): medial temporal atrophy (MTA score), parietal atrophy (Koedam), white-matter disease, microbleeds
- **FDG-PET**: temporo-parietal hypometabolism in AD; frontal in FTD
- **Amyloid-PET**: confirms or excludes Aβ pathology in life
- **Tau-PET**: research and disease-modifying treatment selection
- **DAT scan**: supports DLB when uncertain

Assessment: Biomarkers

- **CSF panel**: Aβ42/40, p-tau181, t-tau (requires lumbar puncture)
- **Plasma p-tau217**: near-CSF accuracy, primary care feasible
- **Plasma Aβ42/40, GFAP, NfL**: pre-screening tools
- **APOE genotyping**: risk stratification; mandatory before anti-amyloid antibodies
- **Genetic testing**: *APP, PSEN1, PSEN2* in early-onset / familial cases

Part 2 — Symptomatic treatments

The development of a field over fifty years, from the cholinergic hypothesis to brexpiprazole.

The cholinergic hypothesis (1976)

- Bowen, Davies, Perry independently observed loss of choline acetyltransferase in post-mortem AD brain
- Source: degenerating cholinergic projections from the **nucleus basalis of Meynert**
- Implication: boost acetylcholine, improve cognition

AChE inhibitor mechanism

Cholinergic synapse cartoon

Presynaptic ACh release → AChE breaking ACh in cleft → AChEi blocking AChE → more ACh available at post-synaptic receptors.

Clinical implications

- AD brains have loss of cortical and hippocampal ACh
- AChEi inhibit synaptic breakdown, increasing transmitter availability
- Effect is presynaptic compensation; does not modify disease
- Side-effects from cholinergic excess: ↑ vagal tone, GI cramping, bradycardia, vivid dreams

AChE inhibitors: clinical use

The three approved AChE inhibitors are presented one per slide.

AChEi: Donepezil

- Once daily, oral; 5 mg → 10 mg after 4 weeks
- Most common UK first-line agent
- Long half-life
- Side effects: GI upset, vivid dreams, bradycardia
- Cardiac history → ECG before starting

AChEi: Rivastigmine

- BD oral or **transdermal patch**
- Patch useful when swallowing or compliance issues
- Inhibits both AChE and butyrylcholinesterase
- Patch reduces GI side-effects substantially
- **Drug of choice in DLB and PDD**

AChEi: Galantamine

- BD oral; also a **nicotinic allosteric modulator**
- Cardiac caution (rare reports of bradyarrhythmia)
- Less commonly used in UK

Memantine: NMDA receptor antagonism

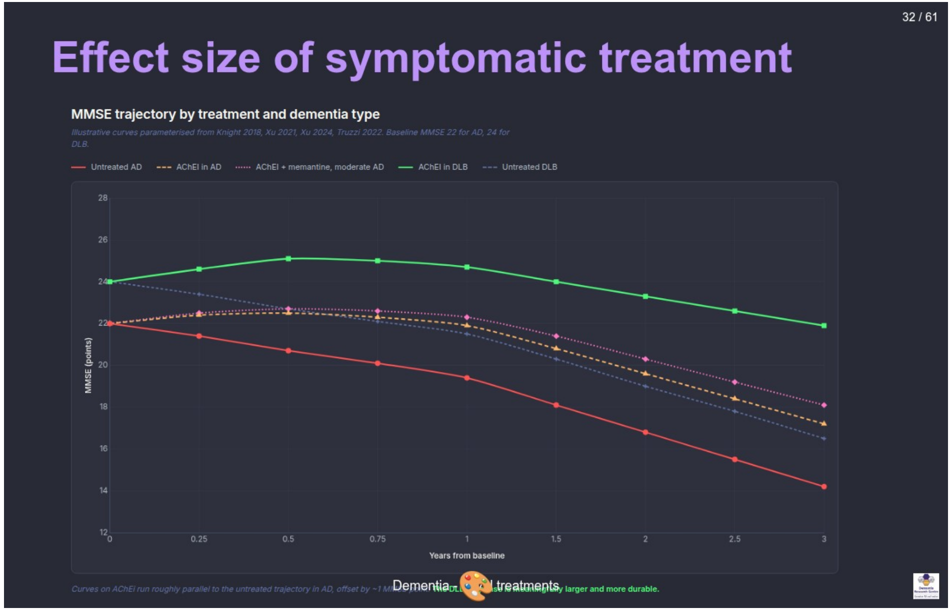
Mechanism

Glutamatergic synapse → NMDA receptor channel → excess glutamate driving pathological tonic activation → memantine sits in the channel as an uncompetitive, low-affinity antagonist.

Clinical use

- Memantine sits in the channel only when pathologically open (voltage-dependent)
- Strong physiological NMDA bursts displace it, preserving learning
- **Indication**: moderate to severe AD, or AChEi-intolerant
- Often combined with donepezil

Effect size of symptomatic treatment



MMSE trajectory over 3 years: untreated AD declines about 2.5 points per year. AChEi shifts the curve up by ~1 point but the trajectory is unchanged. The DLB response to AChEi is much larger.

Differential responsiveness across subtypes

Helpful in

- **AD** (modest, all severities)
- **DLB** (often substantial; rivastigmine first choice)
- **PDD** (cholinergic deficit prominent)
- **VaD** (mixed disease often benefits)

Limited or harmful in

- **bvFTD** (no benefit, may worsen agitation)
- **Pure VaD** (inconsistent)
- **Severe AD** (memantine takes over)

BPSD

- **B**ehavioural and **P**sychological **S**ymptoms of **D**ementia
- Up to 90% of patients experience at least one over the course of illness
- Drives caregiver burden, institutionalisation, and mortality
- Five clusters:
  1. Agitation / aggression
  2. Psychosis (delusions, hallucinations)
  3. Depression / anxiety
  4. Apathy
  5. Sleep disturbance

Non-pharmacological management of BPSD

- **Always rule out a precipitant**: pain, infection (UTI, chest), constipation, drug effect, sensory deprivation
- **Cognitive Stimulation Therapy** (NICE-recommended, group-based)
- **Exercise** (aerobic and resistance)
- **Music therapy**: strong evidence in agitation
- **Environmental modification**: light, routine, signage, familiar objects
- **Caregiver education and support**: often high-yield

Pharmacological management of BPSD

Antipsychotics are last-resort

- ↑ stroke and all-cause mortality (CATIE-AD, FDA black box)
- **Severe sensitivity in DLB**: can cause irreversible parkinsonism / NMS
- Use only when behaviour endangers patient or others, lowest dose, short duration, regular review
- **SSRIs** for depression (citalopram has agitation evidence, but QTc concern)
- **Brexpiprazole**: first FDA approval for agitation in AD, 2023 (not yet NICE-approved on NHS)
- **Trazodone**: sleep, agitation; mostly off-label
- **Avoid benzodiazepines**: falls, paradoxical agitation, dependence

Limits of symptomatic treatment

- They do not slow neuronal loss
- They do not clear Aβ or tau
- They do not change underlying trajectory
- A patient on AChEi 5 years from diagnosis has the same brain pathology as one untreated

Audience poll: what is important in clinical trials?

Interactive Slido poll. Not reproduced in the handout.

5 minute break

Part 3 — Disease-modifying treatments

Anti-amyloid antibodies enter the clinic.

The amyloid hypothesis (Hardy & Higgins, 1992)

- **Aβ accumulation** is the upstream driver of AD
- Genetic anchor: every gene mutation that causes autosomal-dominant AD (*APP, PSEN1, PSEN2*) alters Aβ production or clearance
- Down syndrome (trisomy 21 → 3 copies of *APP*) → near-universal AD pathology by age 50
- Tau pathology, inflammation, neurodegeneration: downstream
- Predicted: lower Aβ → less downstream damage → better outcome

Failed anti-amyloid trials

- **Solanezumab**: Eli Lilly; failed Phase 3 in mild AD (2014, 2016)
- **Bapineuzumab**: Pfizer/J&J; failed (2012); high rate of vasogenic edema
- **Crenezumab**: Roche; failed (2019)
- **Gantenerumab**: Roche; first GRADUATE trials negative (2022); reformulated for prevention

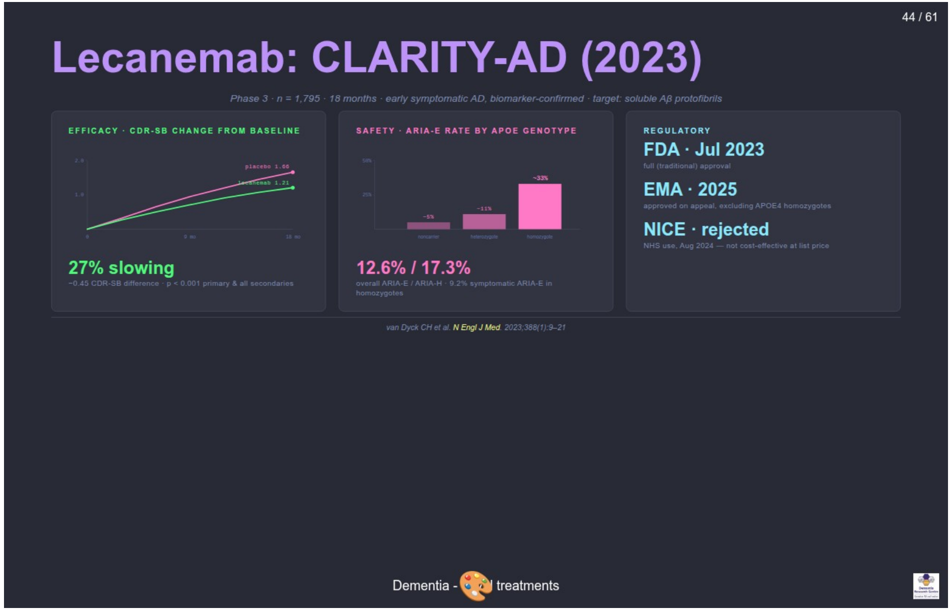
Lessons learned:

- Treating too late (pathology has spread for 15+ years)
- Targeting the wrong species (monomer vs oligomer vs plaque)
- Insufficient brain penetration / dosing
- Endpoint too noisy in mild populations

Aducanumab (2021)

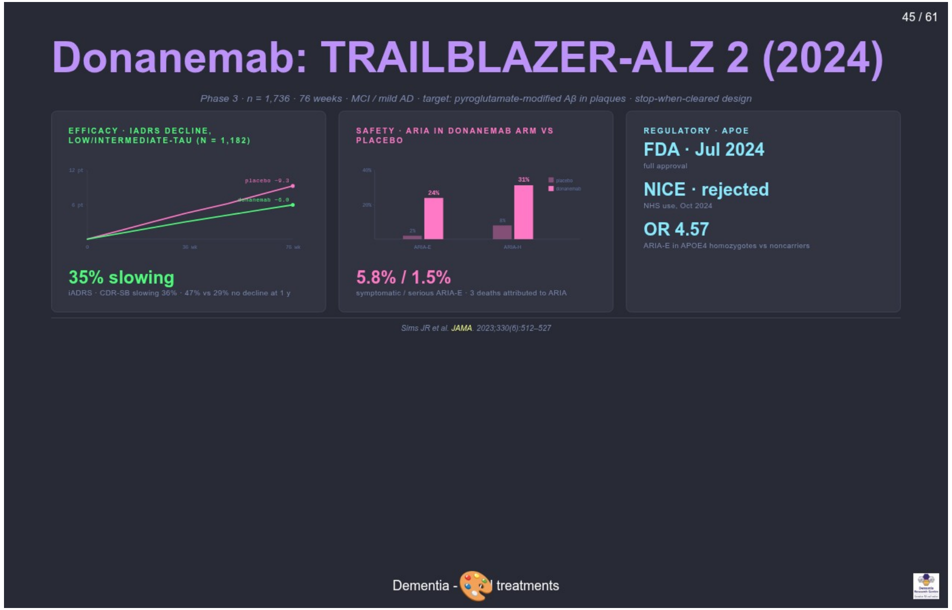
- Two Phase 3 trials (EMERGE, ENGAGE): discordant; pre-specified analysis was negative
- Post-hoc reanalysis at high dose suggested cognitive benefit on EMERGE
- **FDA accelerated approval** June 2021, over near-unanimous advisory panel objection
- EMA **refused** authorisation
- NHS / NICE: **not commissioned**
- Eli Lilly withdrew it from market 2024

Lecanemab: CLARITY-AD (2023)



Lecanemab CLARITY-AD trial summary: efficacy (CDR-SB curves), safety (ARIA by APOE genotype), and regulatory status.

Donanemab: TRAILBLAZER-ALZ 2 (2024)



Donanemab TRAILBLAZER-ALZ 2 trial summary: efficacy (iADRS curves and CDR-SB), safety (ARIA-E and ARIA-H vs placebo), and regulatory.

Antibody specificity

The three anti-amyloid antibodies target different Aβ species along the aggregation pathway:

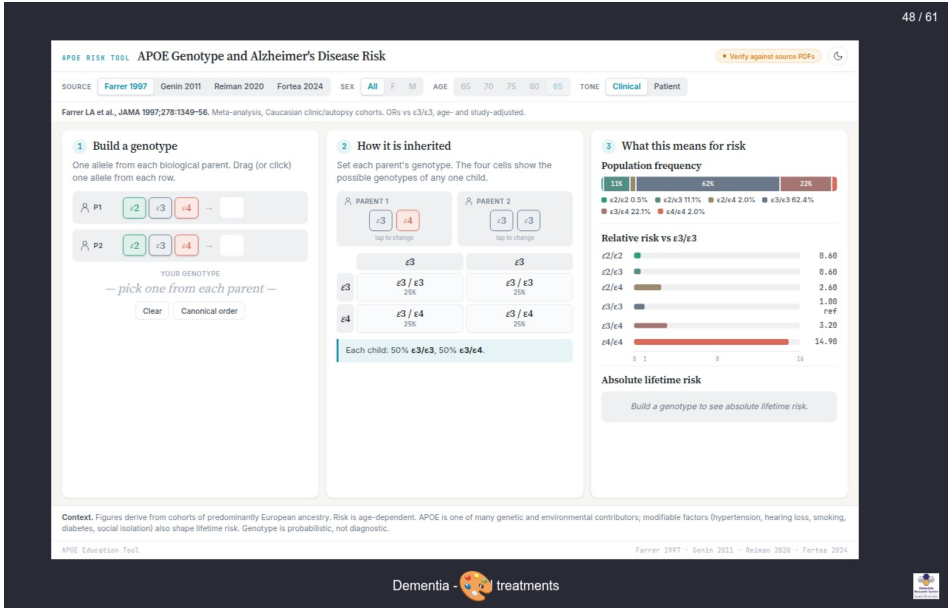
- **Aducanumab**: oligomers + fibrils
- **Lecanemab**: protofibrils selectively
- **Donanemab**: pyroglutamate-modified plaques only

The aggregation pathway runs: monomer → oligomer → protofibril → plaque.

Patient selection

- **Confirmed Aβ pathology**: amyloid-PET or CSF Aβ42/40
- **Disease stage**: MCI or mild AD dementia (not moderate or severe)
- **MRI suitable**: no severe cerebral amyloid angiopathy, ≤4 microbleeds, no superficial siderosis
- **Anticoagulation**: caution or contraindication
- **APOE4 genotyping**: needed for ARIA risk counselling
- **Capable of attending infusions** every 2–4 weeks for 18+ months
- **Capacity** to consent

APOE genotype and ARIA-E risk



ARIA-E risk by APOE genotype for lecanemab and donanemab. Risk rises sharply in heterozygotes and is highest in homozygotes for both drugs.

ARIA

ARIA-E (oedema, effusion)

- Vasogenic oedema, leptomeningeal effusion
- Best seen on **FLAIR**
- Most asymptomatic; caught on surveillance MRI
- Can present with headache, confusion, focal symptoms, seizures

ARIA-H (haemorrhage)

- Microbleeds, superficial siderosis
- Best seen on **SWI / GRE**
- Usually asymptomatic; persists indefinitely after resolution

Severity and surveillance

- Mostly mild; ~3% serious; rare deaths reported
- Surveillance MRI before infusions 5, 7, 14 (lecanemab) or 2, 3, 4, 7 (donanemab)

APOE4 and ARIA risk

APOE genotype	Lecanemab ARIA-E	Donanemab ARIA-E
Non-carrier	~5%	~16%
Heterozygote	~11%	~23%
Homozygote	**~33%**	**~41%**

UK implementation

- **Drug cost**: £20–25k per patient per year (lecanemab list)
- **Infrastructure**: infusion suite every 2 (lecanemab) or 4 (donanemab) weeks
- **Surveillance MRI**: 4–7 scans per patient course
- **Workforce**: trained nurses, neurologists, neuroradiologists
- **NICE August 2024**: rejected lecanemab for NHS commissioning; benefit too small at price
- Currently available only privately in the UK; major equity concern
- Open question whether amyloid lowering alone is sufficient or whether combinations will be needed

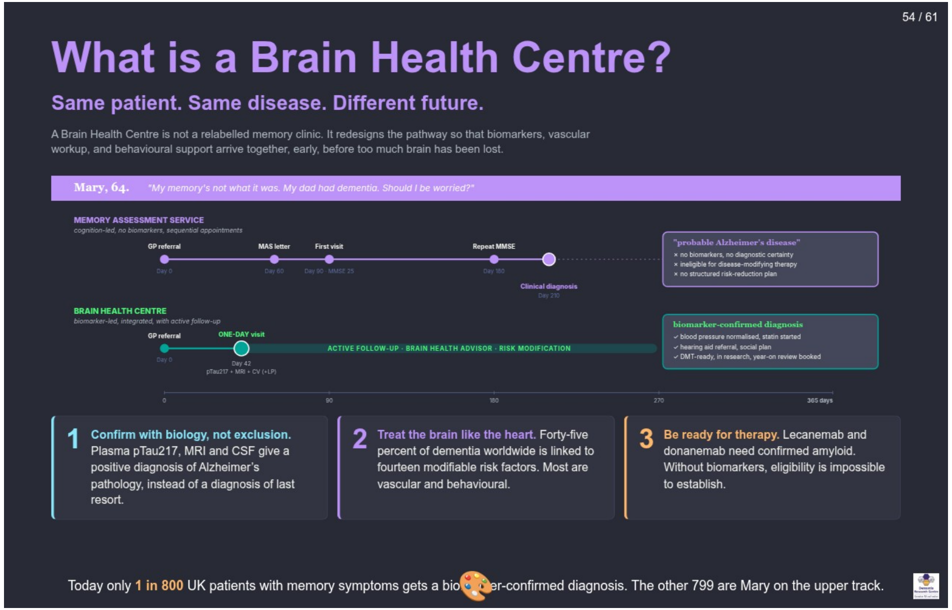
Other approaches

- **Anti-tau antibodies**: E2814, JNJ-2056 (Phase 2; zagotenemab discontinued 2023)
- **Tau ASOs**: intrathecal antisense oligonucleotides, e.g. BIIB080
- **Inflammation**: TREM2 agonists in early trials
- **Vaccines**: UB-311, ABvac40 (Phase 2)
- **Prevention trials** in pre-symptomatic carriers:
  - **DIAN-TU**: autosomal-dominant AD families
  - **AHEAD 3-45**: sporadic, plasma p-tau217 stratified
  - *A4 closed early; solanezumab failed*

Current state

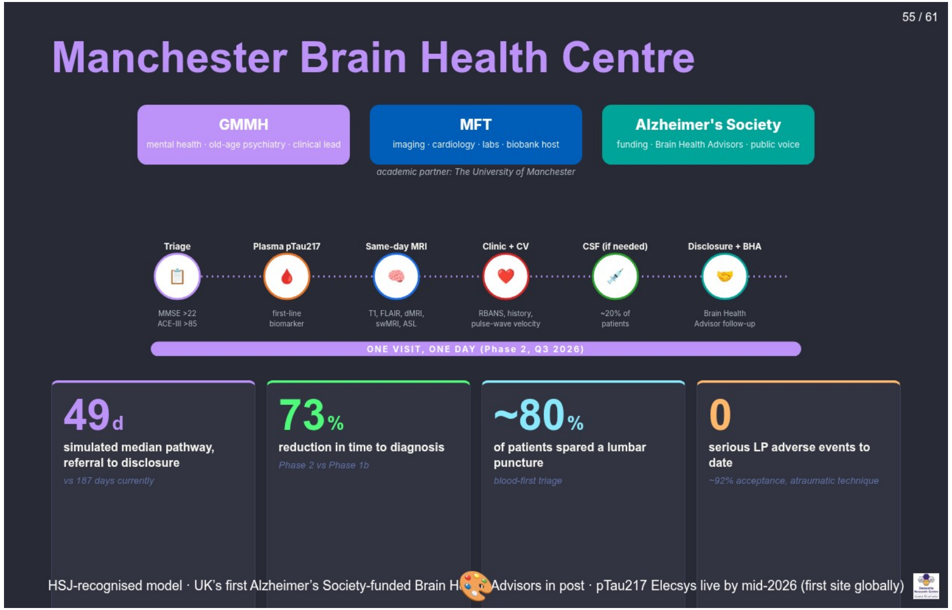
- The first disease-modifying drugs exist. Their effect size is modest. ARIA limits use. Access in the UK is restricted.
- **Diagnosis**: plasma p-tau217 may make primary-care testing feasible.
- **Prevention** may matter more than any antibody:
  - *Lancet Commission 2024*: 14 modifiable risk factors account for ~45% of dementia risk
  - Hearing, education, BP, smoking, obesity, depression, physical inactivity, diabetes, alcohol, social isolation, air pollution, head injury, vision, cholesterol

What is a Brain Health Centre?



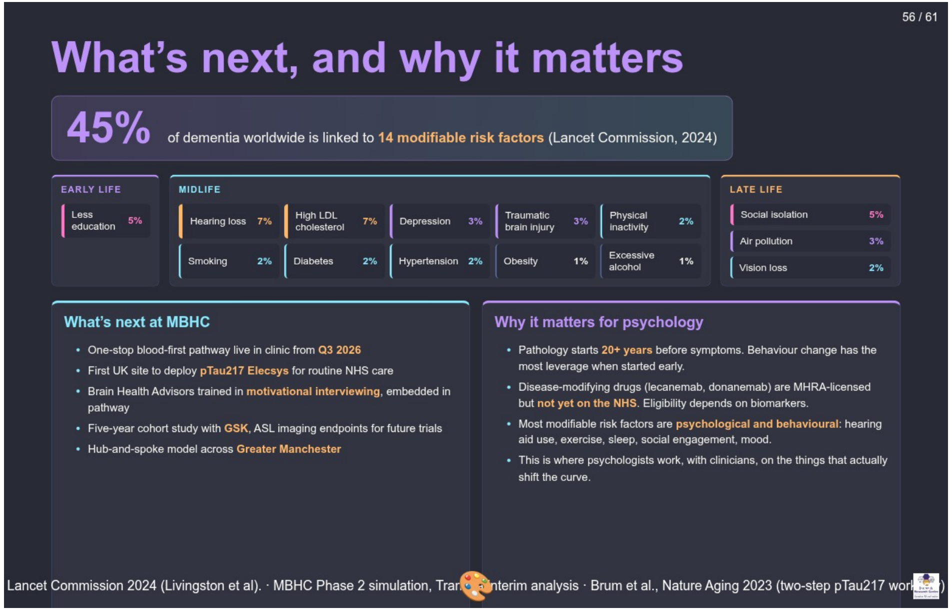
Two parallel pathways for the same hypothetical patient (Mary, 64, presenting with memory concern). The Memory Assessment Service track produces a clinical "probable AD" diagnosis after 210 days, with no biomarkers. The Brain Health Centre track produces a biomarker-confirmed diagnosis in 42 days with a structured risk-reduction plan.

Manchester Brain Health Centre



Three core partners (GMMH, MFT, Alzheimer's Society) plus the University of Manchester. One-day pathway across six stations: triage, plasma p-tau217, same-day MRI, clinic with cardiovascular check, CSF if needed, disclosure with Brain Health Advisor follow-up. Headline KPIs: 49 days median pathway, 73% reduction in time to diagnosis, ~80% spared a lumbar puncture, 0 serious LP adverse events.

What’s next, and why it matters



Headline statistic from Lancet Commission 2024: 45% of dementia worldwide is linked to 14 modifiable risk factors. Risk factors arranged by life stage. Two columns at the bottom: what’s next at MBHC, and why this matters for psychology.

Questions on the material (5 minutes)

Open floor.

Q&A with our trial participant (10–20 minutes)

Welcome.

Student questions (10 minutes)

Over to you.

Summary

1. **Dementia is a syndrome; AD is defined biologically.** Aβ and tau can be detected in life via CSF, plasma p-tau217, or PET.
2. **Symptomatic treatments help modestly and do not change disease course.** AChEi and memantine offer about a 6-month delay. Non-pharmacological BPSD interventions are first-line. Antipsychotics carry mortality risk.
3. **Anti-amyloid antibodies are the first disease-modifying drugs.** Effect size is modest. ARIA limits use. NICE has rejected them for the NHS. Prevention may matter more.

References (1/2)

- Bowen DM, Smith CB, White P, Davison AN. Neurotransmitter-related enzymes and indices of hypoxia in senile dementia and other abiotrophies. *Brain* 1976; 99(3): 459–96.
- Hardy JA, Higgins GA. Alzheimer's disease: the amyloid cascade hypothesis. *Science* 1992; 256(5054): 184–5.
- Jack CR Jr, Bennett DA, Blennow K, et al. NIA-AA Research Framework: Toward a biological definition of Alzheimer's disease. *Alzheimer's & Dementia* 2018; 14(4): 535–62.
- Livingston G, Huntley J, Liu KY, et al. Dementia prevention, intervention, and care: 2024 report of the Lancet standing Commission. *The Lancet* 2024; 404(10452): 572–628.

References (2/2)

- Palmqvist S, Tideman P, Mattsson-Carlgren N, et al. Blood biomarkers to detect Alzheimer disease in primary care and secondary care. *JAMA* 2024; 332(15): 1245–57.
- Sims JR, Zimmer JA, Evans CD, et al. Donanemab in early symptomatic Alzheimer's disease: the TRAILBLAZER-ALZ 2 randomized clinical trial. *JAMA* 2023; 330(6): 512–27.
- van Dyck CH, Swanson CJ, Aisen P, et al. Lecanemab in early Alzheimer's disease. *New England Journal of Medicine* 2023; 388(1): 9–21.